



STRAT*EX*

PROJECT AD-KY041 • 142 Bluegrass Ridge • Lexington, KY

CONFIDENTIAL • STRATEGIC FORENSIC AUDIT



STRAT*EX*TM

FORENSIC ENVELOPE AUDIT • V1.0

Strategic Thermal Reconnaissance Report

142 Bluegrass Ridge • Lexington, KY • Scan 2026-06-24 • Ground-truth accuracy ±1.5 cm

TOTAL SQUARES

26.80

SQ (100 sf)

SHINGLE LAYERS

1 / 2

DETECTED / CODE MAX

FACETS

6

DISTINCT PLANES

ANOMALIES FLAGGED

4

FORENSIC HOTSPOTS

EXECUTIVE SUMMARY • GRAND TOTAL RANGE

\$12,082.29 → \$13,232.29

MATERIALS \$6,474.29 • LABOR \$3,350.00 • TEAR-OFF \$1,608.00 • UNFORESEEN \$650.00-\$1,800.00

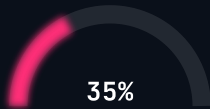
STRATEX RECOMMENDATION

TEAR OFF

Roof is 19 years old, showing signs of end-of-life wear and granule loss. Current shingle layers detected is 1, so an overlay is technically possible, but a tear-off provides the opportunity for a thorough deck inspection and ensures optimal performance of a new roofing system.

Building Envelope Score

ENERGY DEFICIENCY



DEFICIENT

VENTILATION



POOR

OVERALL ENVELOPE

60/100

COMPOSITE

AI-PRIORITIZED MAINTENANCE QUEUE

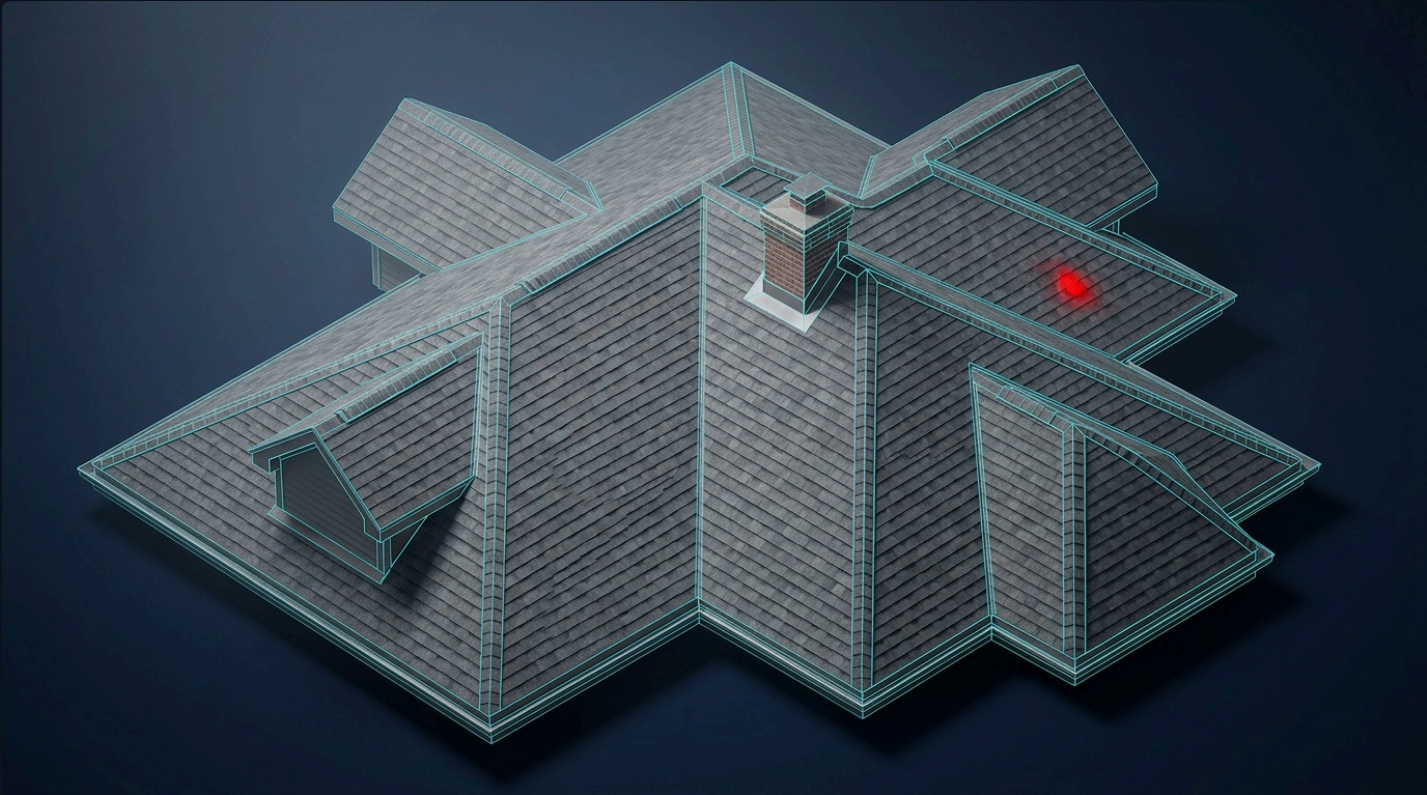
1	URGENT	Repair/Replace Plumbing Vent Boot (A002) to prevent water intrusion.	\$0/yr
2	HIGH	Full Roof Replacement (TEAR_OFF) due to age and widespread wear.	\$0/yr
3	HIGH	Address Attic Heat Loss (Thermal Finding) and seal Attic Access Hatch (Energy Leakage).	\$138/yr
4	MED	Replace exterior door weatherstripping, especially rear patio door (D2).	\$110/yr
5	MED	Caulk/seal window perimeters (W1, W4) to reduce drafts.	\$165/yr

THERMAL FORENSIC FINDINGS

Attic Heat Loss	15°C above ambient on roof ridge and vents	HIGH
Window Perimeter Leakage	2-3°C temperature differential at window frames	MED
Door Threshold Draft	3-4°C temperature differential at main entry door bottom	HIGH

Sub-surface Anomaly Atlas

RADIOMETRIC OVERLAY • ROOF



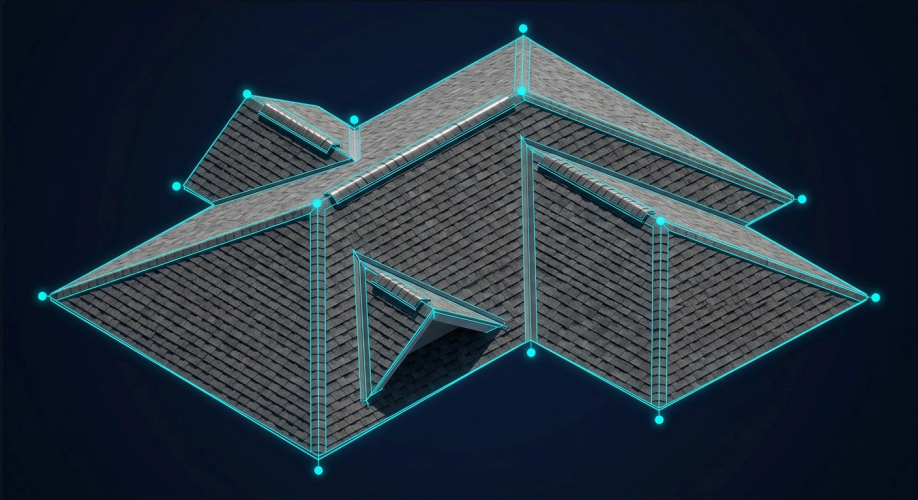
CYAN: scan-vector overlay • ORANGE: thermal hotspot • Confidence threshold ≥ 85%

ANOMALY LEDGER

SEV	ID	TYPE	LOCATION	CONF	SF	DIAGNOSIS	EST
MED	A001	Granule Loss	Front-right slope	95.0%	4.7	Typical aging, indicative of advanced wear on south-facing slopes. Affects shingle integrity.	\$150.00
HIGH	A002	Vent Boot Seal Failure	Main plumbing stack vent, rear slope	98.0%	0.1	Deteriorated rubber seal, high potential for water intrusion into attic space.	\$350.00
MED	A003	Curled Shingles	Various sections, predominantly south-facing	90.0%	9.3	Age-related thermal cycling and moisture absorption, reducing wind resistance and weatherproofing.	\$200.00
LOW	A004	Missing Gutter Hanger	Front gutter, 3.5m from left corner	99.0%	0.0	One gutter hanger detached, causing minor sag. Risk of increased sag and water overflow if not addressed.	\$75.00

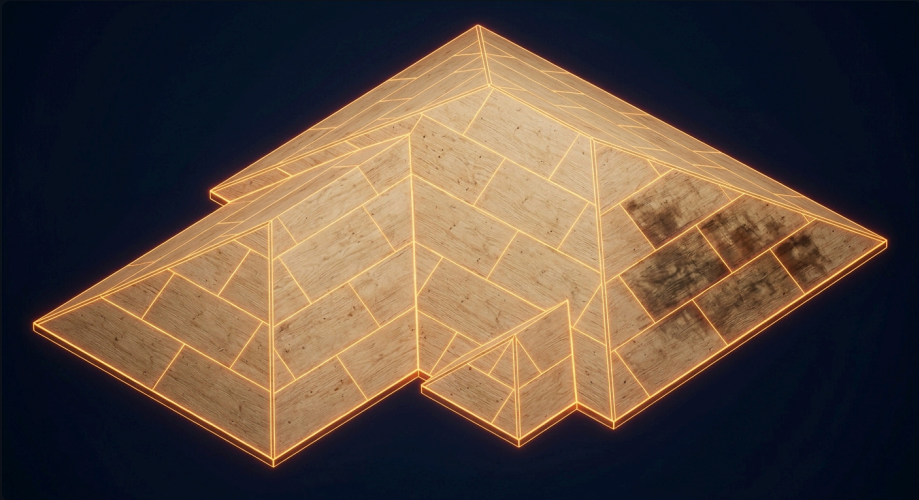
Layered CAD/BIM Twin

LAYER 01 • FINISHED SLATE



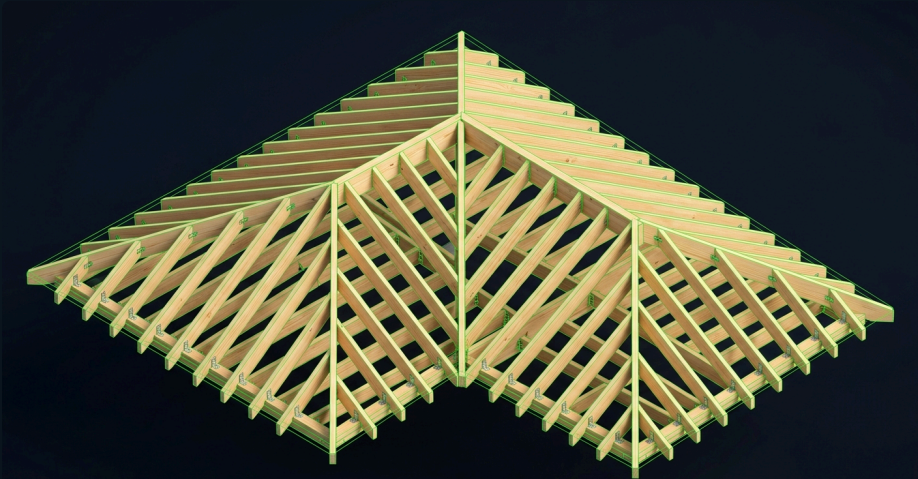
Squares 26.80 • Pitch 8/12

LAYER 02 • DECKING + UNDERLAYMENT



7/16" CDX • I&W shield at valleys/eaves • 1 existing shingle layer(s)

LAYER 03 • STRUCTURAL FRAMING



2x8 rafters @ 16" o.c. • LVL ridge • Hurricane ties **VERIFIED**

VALLEYS

24.4

LINEAR FT

GABLES

30.5

LINEAR FT

RIDGES

18.3

LINEAR FT

EAVES

45.7

LINEAR FT

Fenestration Schedule

ID	LOCATION	SHAPE	W	H	QTY	U-FACTOR	LEAK SEV
W1	Front Elevation	Double-Hung	36"	60"	3	0.35	MED
W2	Side Elevation (Left)	Double-Hung	28"	48"	2	0.35	LOW
W3	Side Elevation (Right)	Casement	24"	48"	1	0.32	NONE
W4	Rear Elevation	Picture w/ Flanking Double-Hung	72"	60"	1	0.33	MED

Exterior Door Schedule

ID	LOCATION	SHAPE	W	H	QTY	WEATHERSTRIP
D1	Front Entry	Single Panel w/ Sidelights	36"	80"	1	WORN
D2	Rear Patio	Sliding Glass	72"	80"	1	FAILED

• WALL ENVELOPE & SIDING PACKAGE • GEOMETRY + MATERIAL AGENT

Vertical Envelope + Vinyl Package

GROSS WALL SF

214

PRE-SUBTRACTION

FENESTRATION

9

WINDOWS + DOORS

NET WALL SF

205

SIDING TARGET

WALL HEIGHT

9.0'

EAVE-TO-GRADE

ELEVATION BREAKDOWN

ELEV.	GROSS SF	NET SF	MOISTURE
Front	55.7	50.2	5%
Rear	55.7	48.1	8%
Left	51.1	49.3	3%
Right	51.1	47.3	2%

SIDING PACKAGE • Vinyl

PROFILE: Dutch Lap • COLOR: Light Beige

ACCESSORY	QTY	UNIT	UNIT	LINE
J-Channel	18	12ft pcs	\$8.00	\$144.00
Outside Corner Post	4	ea	\$25.00	\$100.00
Window/Door Lineals	12	12ft pcs	\$15.00	\$180.00
SIDING SUBTOTAL				\$424.00

• WALL MOISTURE + VAPOR BARRIER • THERMAL AGENT

Sub-SidingMoisture Atlas

VAPOR BARRIER STATUS

OK

BARRIER PRESENT • INFRARED-CONFIRMED

ZONES FLAGGED

1

THERMAL ANOMALY ZONES • WALL CAVITY

REMEDIATION TOTAL

\$450.00

ROLLED INTO MATERIALS TOTAL

THERMAL SATURATION ZONES • WALL CAVITY

ID	ELEV.	AREA SF	MOISTURE	DAMAGE PROB.	REMEDIATION	EST.
MZ001	Rear	0.5	18%	60%	Seal small crack in siding, replace localized insulation, repair interior drywall	\$450.00

Envelope Air-Leak Profile

BLOWER-DOOR ACH₅₀ (EST.)

7.5

TARGET ≤ 3.0 • LEAKY ENVELOPE

ANNUAL ENERGY LOSS

25,000 kBTU

SEASONAL LOSS • CONDITIONED ENVELOPE

ANNUAL DOLLAR LOSS

\$550.00 /yr

RECOVERABLE WITH REMEDIATION

LEAK-SOURCE LEDGER

SOURCE	LOCATION	BTU LOSS	\$/YR	RECOMMENDED FIX
Attic Access Hatch	Hallway ceiling	25%	\$138/yr	Weatherstrip and insulate access hatch
Window Perimeter Gaps	Various windows	30%	\$165/yr	Caulk exterior gaps, replace interior weatherstripping
Exterior Door Seals	Front and Rear doors	20%	\$110/yr	Replace door sweeps and weatherstripping
Electrical Outlets/Switches	Exterior walls	10%	\$55/yr	Install foam gaskets behind plates

Bill of Materials

CATEGORY	MATERIAL	SKU	QTY	UNIT	UNIT COST	LINE TOTAL
Shingles	Architectural Shingles	CertainTeed Landmark Pro	26.8	squares	\$125.00	\$3,350.00
Underlayment	Synthetic Underlayment	Titanium UDL30	27	squares	\$25.00	\$675.00
Ice & Water Shield	Ice & Water Shield	GAF LeakBarrier	3	rolls	\$80.00	\$240.00
Vents	Ridge Vent	GAF Cobra Snow Country Advanced	18.29	lf	\$8.50	\$155.47
Vents	Plumbing Stack Boot	Perma-Boot 1.5-3in	3	ea	\$20.00	\$60.00
Flashing	Drip Edge	Aluminum Drip Edge	45.72	lf	\$2.50	\$114.30
Flashing	Valley Flashing	W-Style Valley Metal	24.38	lf	\$4.00	\$97.52
Fasteners	Roofing Nails	1.25in Coil Nails	8	boxes	\$30.00	\$240.00
Disposal	Dumpster Fee	20-yard container	1	ea	\$500.00	\$500.00
Accessories	Starter Shingles	GAF Pro-Start	6	bundles	\$28.00	\$168.00
MATERIALS TOTAL						\$6,474.29

Crew Economics

RATE BENCHMARK

NATIONAL AVG / SQ

\$130.00

REGIONAL AVG / SQ

\$120.00

REGION: Midwest (Kentucky) • APPLIED: \$125.00/SQ

CREW PLAN

CREW SIZE

5

DAYS

2.5

LABOR TOTAL

\$3,350.00

Unforeseen Repair Reserve

ITEM	PROB	LOW	HIGH	NOTE
Plywood Sheathing Replacement	20%	\$300.00	\$800.00	Possible water damage or rot on roof deck, typically discovered during tear-off.
Fascia/Soffit Repair	15%	\$200.00	\$600.00	Minor rot or damage due to previous water egress, requires repair before new drip edge/gutters.
Gutter Replacement (Partial)	10%	\$150.00	\$400.00	Localized damage beyond repair or upgrade requested.
RESERVE RANGE		\$650.00	\$1,800.00	

Full **Tear-Off + New Install** Required

WATER RETENTION PROBABILITY

10%

DEPTH EST 0.00" • MULTI-SEASON ENTRAPMENT

R A T I O N A L E

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TEAR-OFF COST INCLUDED

\$1,608.00